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Grand Unification Theories And Supersymmetry

Grand Unification Theories (GUTs) and supersymmetry are advanced frameworks in theoretical physics that aim to unify the fundamental forces of nature into a cohesive model. GUTs strive to combine the strong, weak, and electromagnetic forces, while supersymmetry seeks to incorporate gravity into this unified schema. At the core of these theories is the idea that these forces, although manifesting differently, share underlying similarities that can be described mathematically.

The four fundamental forces include gravity, which is the attraction between massive bodies; electromagnetism, which governs interactions between charged particles; and the strong and weak nuclear forces, which operate primarily within atomic nuclei. The quest for grand unification culminates in the concept that at extremely high energies, these forces may converge, revealing a common framework. Supersymmetry introduces a relationship between bosons and fermions, suggesting that each known particle has a superpartner, thereby enriching the particle spectrum.

Predictions from GUTs and supersymmetry have inspired numerous experimental investigations, including searches for proton decay and dark matter, as they may shed light on unsolved mysteries in cosmology and particle physics. Although these theories have not yet been confirmed, they represent significant advancements in understanding the universe's fundamental structure and evolution.

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Type of physical science: Elementary particle (high-energy) physics

Field of study: Unified theories

Theories that encompass all or most of the forces of nature are said to be unified theories. Grand unified theories attempt to unite physicists' knowledge of the strong, weak, and electromagnetic interactions. Supersymmetry promises to include gravity in this single scheme.

Overview

Grand unification theories and supersymmetry are two classes of theories that attempt to unify most of the knowledge concerning the forces of nature into a single structure, consistent with [quantum mechanics](#) and relativity. These theories build on the common similarities among the forces but must be flexible enough to allow the obvious differences to be properly described.

There are four forces whose manifestations are sufficiently different as to be separately identified. The most familiar is the force of gravity. Gravitation is the attractive force between all massive bodies. As a result of gravity, an object released from above the earth's surface will fall to the ground. The electromagnetic force is responsible for a variety of common phenomena, such as radio waves and light. One of its basic manifestations is the electrostatic force between charged objects, such as the "static cling" in clothes dryers. Electromagnetic forces hold atoms and molecules together. Finally, the strong and weak interactions have a very short range, on the order of 10^{-12} of a millimeter, and so are only operative in the realm of nuclear physics. These two forces are distinguished by their relative strength and by their selection rules. Selection rules restrict the types of reactions that can occur. For example, the fact that the charge of particles interacting electromagnetically does not change would be considered a selection rule of the electromagnetic interaction.

[Electromagnetism](#) is the best understood of the four forces. The classical understanding of electromagnetism emerged during the nineteenth century as various phenomena, such as magnetic attraction, electric currents, radio waves, and optics, were all found to be different sides of the same fundamental force. (Electromagnetism was the first unified theory and led directly to Albert Einstein's theory of relativity.) Much has been learned by analogy with

electromagnetism.

For example, the weak interaction has much in common with electromagnetism. The "standard model" of elementary particles includes a unified theory of weak and electromagnetic interactions, as well as a theory of the strong interaction which is similar to electromagnetism in many ways. The theory of the strong interaction is based on a model of quarks and gluons and is known as [quantum chromodynamics](#) (QCD). The word "chromo" here refers to "color." "Color" has nothing to do with what quarks and gluons look like; rather, it is a kind of charge, like an electric charge. The "color" of the strong interaction comes in six varieties. Each quark has a color and each gluon has two colors. This multiplicity of charge is what gives the strong force its unique properties.

In general, each of the fundamental forces of nature is thought to be mediated by a class of particles called bosons. For electromagnetism, this boson is the photon. The photon is the particle manifestation of the electromagnetic force. Many properties of the photon can be inferred from easily observed properties of light and radio waves. In particular, it is known that light and radio waves can be polarized; that is, the electric field can have a specific orientation.

This reflects an internal degree of freedom in the photon, which is known as spin. Spin refers to a particle's intrinsic angular momentum. The photon is a massless "spin 1 boson." Scientists know it is massless because the range of electromagnetic forces is infinite. While the force drops with distance, this drop is only as the square of the separation distance, a purely geometric effect.

Careful studies of weak nuclear decays, such as β decay, indicate that the weak interaction is also mediated by a spin 1 boson. This is the major similarity with electromagnetism, but there are numerous obvious differences. The weak interaction has a very short range, on the order of the size of elementary particles. Weak interactions always seem to involve a change in charge of the interacting particles; electromagnetism never does. In spite of these differences, the standard model has succeeded in combining these two interactions. One manifestation of this unification is the weak neutral current, which, while still of a short range, has selection rules that are similar to electromagnetism.

The standard theory of the strong nuclear force is fairly complicated. The force that binds [protons](#) and neutrons inside the nucleus is considered to be an indirect manifestation of this interaction. The strong force's primary role is to bind quarks to form elementary particles. The proton and neutron are each made up of three quarks.

The quarks are bound together by spin 1 bosons called gluons. Most of the knowledge of the structure of proton and quantum chromodynamics is indirect. These structures have been studied using electron scattering as a probe. This scattering reveals that the internal structure of the proton must contain some components that are uncharged and do not interact with

electrons.

These components are the gluons.

Gluons are uncharged and massless, and therefore they are very similar to the photon of electromagnetism. The most obvious difference is that the strong force is of a very short range, perhaps because of an effect called color screening. The strong charge comes in six "colors," but scientists have never observed particles with these color charges because the confinement force requires all observable objects to be color-neutral. At long distances, the color force becomes very strong and the force is shielded by the production of additional colored particles from the vacuum. The net color of all of these must be neutral. The similarity between the strong force and electromagnetism is most pronounced at very short distances when the colored force becomes very weak. This effect of weakening at very short distances was discovered, theoretically, in the early 1970's and has been given the name "asymptotic freedom."

The similarities among these three forces of nature, however, are obscured by their obvious differences. Electromagnetic interactions are of infinite range, while the strong and weak interactions are of a very short range. Furthermore, the reasons for this short range are different for strong and weak interactions. The complementary feature to asymptotic freedom is that the strong interaction becomes very strong at large distances. The forces become so large that influenced particles cannot move very far apart. Therefore, the leading manifestations of this force are masked, since all physical systems are color-neutral and have only indirect strong interactions. If much energy is used to try to overcome the confining force, instead of separating particles to large distances, then more color-neutral particles are produced. The weak interaction, on the other hand, is of short range because the interaction is mediated by particles of very large mass. There is no confinement mechanism, as there is for QCD, and the spin 1 bosons that are associated with the weak force have been produced and observed in the laboratory.

The similarities and differences that typify the strong, weak, and electromagnetic interactions make it necessary that any theory that "unifies" them, or accounts for all these forces, must also permit them to manifest their obvious differences. Hence, one would say that the symmetry that unites them is "broken" to provide the forces that are manifest in the world. The world is of relatively low energy, so any unification is expected to be manifested only at very high energies. At high energies, calculations involving asymptotic freedom reveal that the strong interaction becomes weak. Similar calculations indicate that the weak and electromagnetic interactions become stronger. The weak and electromagnetic interactions achieve a partial unification at an energy that is about one hundred times the mass of the proton. The unification of these two forces with the strong interaction, which occurs at about 10^{15} times the mass of the proton, is called a "grand unification."

The fourth force--gravitation--is not yet included in grand unification schemes. Unlike the other

three forces, gravitation is not mediated by a spin 1 boson. Instead, the structure of the theory of gravitation indicates that gravity is mediated by a massless spin 2 boson known as a "graviton." To include the graviton in a unification scheme would require a theory that mixes particles of different spins. Such a theory has been developed, and the class of theories is known as "supersymmetry." Supersymmetric theories that include gravity are sometimes given the name "supergravity." Supersymmetric theories have some very nice properties. The great degree of symmetry in the models helps control some of the problems that come about when one attempts to produce a quantum theory of gravity. Supersymmetry cannot be an exact symmetry of nature, since supersymmetry predicts that all particles would have the same mass, which is not the case.

Theories with broken supersymmetry may not be able to retain the good features of unbroken supersymmetry and to be consistent with the known particle spectrum.

The price that supersymmetry must pay for including the force of gravity is that it must also include the large numbers of new particles that occur because the theory needs many degrees of freedom to accommodate the unique features of gravity. After the unification is accomplished, there are still many particles left over. The lightest of these should be stable. An active search is under way for this particle, which may be labeled the photino or the neutralino.

Applications

Unified theories are based on the common features of those subjects to be unified. A unified theory must be more complex than its constituent theories, since it must incorporate the various constituent theories. Yet, a unified theory is more than an enumeration of the particular subtheories. In particular, the structure of the unified theory usually encompasses more than its constituents. Because of this, the unified theory can use information from one area to make predictions in another, and it often will contain phenomena that none of the constituent theories possesses. One of the early successes of grand unification was the prediction of the strength of the weak neutral current through a calculation that involved quarks. This prediction of a weak interaction parameter using strong interaction data was highly accurate.

Grand unified theories must account for some very weak interactions that have not yet been discovered. Basically, it is not possible to construct a grand unified theory without including at least twelve additional spin 1 bosons. It is expected that the masses of these bosons would be very large, near the unification mass scale, so that any interactions they would cause would be very weak. Fortunately, it is easy to predict the existence of even very weak interactions if they produce an effect that could not occur without their presence. The gauge bosons of grand unification would permit protons to decay.

Protons and neutrons, the stuff of ordinary matter, are so stable that, in the past, people have

proposed conservation laws to explain their stability. It is now thought that protons are only approximately stable and that they would eventually decay to electrons and photons. The predicted lifetime of a proton is very long, in excess of a 10^{27} years, but such incredibly long times are still within range of experimental study. Instead of waiting this long, one can look at a 10^{27} protons for only a few years in order to determine whether they decay. Such experiments have been carried out and indicate that the lifetime must be at least one thousand times longer than that originally predicted. Furthermore, no experiment can conclude that such an effect never occurs. The best that one can hope to do experimentally is to set a very restrictive boundary on the decay processes.

These limits on proton decay have, in fact, ruled out the simplest of the grand unification models, but many more complicated models exist, and their predictions are less specific or harder to test. Some models indicate that the neutron is able to convert to an antineutron, a process which could occur even if protons do not decay. A number of experiments have looked for this neutron-antineutron oscillation with free neutrons. The search is difficult, however, because the free neutron itself is an unstable particle. Experiments that have placed limits on the lifetime of the proton have also been able to look at this neutron oscillation effect in neutrons bound in the matter containing the protons. For bound neutrons, the effect is reduced by several orders of magnitude, but the stability of the bound neutron makes this approach competitive with those done with free neutrons.

All grand unification schemes that incorporate electromagnetism predict the existence of magnetic monopoles. A magnetic monopole is a particle which carries a magnetic charge. A magnetic charge is the source of magnetic fields in the same way that an electric charge is the source of electric fields. It is known that magnetic fields are produced by moving electrically charged particles, but the magnetic field of a monopole would have a very different field configuration. Magnetic lines of force would begin or end on the magnetic charge. Magnetic charges are not needed to understand electromagnetism, but they can be accommodated and would provide a symmetry between magnetism and electricity that a theory without them lacks.

Grand unification predicts that the monopole mass would be comparable to the unification mass.

It is also believed that magnetic monopoles could "catalyze" proton decay--that is, in the presence of a monopole, protons would decay at a rate typical of strong interactions. In spite of extensive searches, however, no one has ever reported a confirmed observation of magnetic monopoles.

Grand unified theories are very important for understanding the early universe. As one goes back in time closer and closer to the big bang, the energy density rises. Consequently, the energies of interactions go up and the unification scale is approached. A number of important questions, such as the excess of matter over antimatter in the universe, can be addressed in the

context of grand unified theories.

Supersymmetry has predicted a new particle spectrum which is similar to the known particle spectrum. These new particles should be found in nuclear reactions, unless they are very heavy. Many attempts to produce them have been made, but no conclusive signals have been seen at accelerators. Indirect evidence has been sought in both rare particle decays and cosmology. Just as proton decay is predicted in grand unified theories, the existence of supersymmetric matter would permit very rare particle decays. Observation of these decays would be indirect evidence for the theory of supersymmetry.

If supersymmetry is correct, then its particles would have been produced in the big bang. Some of them would remain today. These particles may be responsible for the formation of galaxies and the distribution of normal matter in the universe. Indirect evidence for them can be sought, since they will collide with astronomical objects and perhaps be captured.

Supersymmetric particles may make up the missing mass, the dark matter, of the universe.

Searches to observe dark matter by its capture or decay in the universe have so far only placed limits on its existence. Efforts are under way to observe dark matter directly, in the laboratory.

Context

Grand unified theories are the logical next step in an understanding of the universe as a whole. These theories emerged after the success of the "standard model" of elementary particles, which includes a unified theory of electromagnetic and weak interactions. The obvious similarities among the major forces of nature made a grand unified theory a reasonable objective.

Abdus Salam, in 1973, was the first to propose uniting leptons, which do not interact strongly, with quarks that do. Howard Georgi and Sheldon L. Glashow made the concept consistent with the standard model. Their model predicted parameters which the standard model required but could not predict. It even gave a calculable lifetime to the proton. Searches were initiated in the late 1970's to test this prediction. This simplest grand unification scheme has been found to be in conflict with direct observation. More complicated schemes exist but do not make unique predictions. Grand unification theories provide a motivation for experiments searching for proton decay, magnetic monopoles, and dark matter. The beauty of these grand unified theories is that many diverse phenomena could be collected under one theory and many seemingly disparate phenomena could be understood as different manifestations of one force.

Supersymmetry emerged in 1976 as an attempt to include the entire particle spectrum, both bosons (1 spin) and fermions (1 1/2 spins), in a unified way. Many theoretical problems still exist, but the theory has evolved, through the work of John H. Schwarz and Michael Green in the mid-1980's, into the modern theory of superstrings.

Grand unification has provided a context for much experimental and theoretical work beyond the standard model of elementary particles. While grand unification has not been verified, it has provided a very fruitful inspiration for creative ideas about why the forces of nature share so many similarities. Supersymmetry has inspired a considerable amount of thought about the natures of space, time, and gravity. The most significant predictions of supersymmetry are the existence of a vast spectrum of particles that are the superpartners of the ones that are already known. Some of these particles may play a major role in cosmology and the shaping of the universe.

Principal terms

ASYMPTOTIC FREEDOM: the process that permits the strong interaction to become weaker as the energy of a reaction rises

CONFINEMENT: the process that prevents quarks and gluons from leaving protons, neutrons, and other particles; another aspect of asymptotic freedom

ELECTROMAGNETIC INTERACTIONS: the class of interactions between electrically charged particles; these interactions involve the photon, which is associated with the electromagnetic field

GRAVITATION: the force between large massive bodies that governs the motion of the planets and the stars; while the weakest known force, it is the most significant on a large scale because the force of gravity cannot be shielded or its "charge" neutralized

QUARKS: the fundamental building blocks of protons, neutrons, and other strongly interacting particles

SPIN: the intrinsic angular momentum of a particle on which many properties depend; its value is restricted to be a multiple of a fundamental quantum, which is measured in units of Planck's constant divided by 2π

STRONG INTERACTION: the force that binds the atomic nucleus together because it is stronger than the electromagnetic repulsion between protons; the most obvious manifestation of the strong force, which is also believed to hold the constituents of the proton and neutron together

SYMMETRY: the property of remaining unchanged after an operation has been performed

WEAK INTERACTION: the force that is responsible for the decay of many particles, including nuclear β decay; it is far weaker than the electromagnetic force but not as weak as gravity

Bibliography

Carrigan, Richard A., Jr., and W. Peter Trower, eds. *PARTICLE PHYSICS IN THE COSMOS*, New York:

W. H. Freeman, 1989. A collection of reprints from SCIENTIFIC AMERICAN on particle physics. Several articles examine grand unified theories, and many present the motivation for and the results of experiments attempting to confirm the theory.

Crease, Robert P., and Charles C. Mann. THE SECOND CREATION: MAKERS OF THE REVOLUTION IN TWENTIETH CENTURY PHYSICS. New York: Macmillan, 1986. This book concentrates on the history and personalities involved in the development of unified theories. The excitement of discovery can be sensed throughout its pages.

Davies, Paul C. W. SUPERFORCE: THE SEARCH FOR A GRAND UNIFIED THEORY OF NATURE. New York: Simon & Schuster, 1984. A general, nonmathematical introduction to grand unification and supersymmetry. Davies includes an introduction to quantum mechanics and relativity. Applications of grand unified theories and supersymmetry, including cosmology, are discussed in great detail.

Georgi, Howard. "A Unified Theory of Elementary Particles and Forces." SCIENTIFIC AMERICAN 244 (April, 1981): 48-63. This article goes into great detail about the structure of the standard model of elementary particles and the common features of the known interactions. After laying this foundation, it goes on to explain why grand unification is possible and the consequences that it would have. The simplest model, based on the symmetry SU, is emphasized.

Haber, Howard E. and Gordon L. Kane. "Is Nature Supersymmetric?" SCIENTIFIC AMERICAN 254 (June, 1986): 52-61. This article discusses the motivation behind the search for supersymmetry and some of the consequences of such a unification scheme. Argues that conventional experiments, such as rare particle decays, are possible to explore supersymmetric interactions.

LoSecco, J. M., Fredrick Reines, and Daniel Sinclair. "The Search for Proton Decay." SCIENTIFIC AMERICAN 252 (June, 1985): 54-63. This article discusses some of the experimental methods used to study grand unified theories. The use of massive detectors, which are heavily shielded from cosmic rays and other forms of natural radioactivity, is described. Such methods are needed to access the very long decay times that are expected in these theories.

Parker, Barry. SEARCH FOR A SUPERTHEORY. New York: Plenum Press, 1987. A historical overview of modern particle physics. Most of the emphasis is on the standard model, but later chapters discuss the evolution of grand unified and supersymmetric theories.

Weinberg, Steven. "The Decay of the Proton." SCIENTIFIC AMERICAN 244 (June, 1981): 64-76. In this article, Weinberg examines the question of the stability of ordinary matter. He explains conservation laws and looks at baryon (proton) conservation in the light of grand unification.

Quarks and the Strong Interaction

The Unification of the Weak and Electromagnetic Interactions

Essay by John M. LoSecco

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